The Fiscal Incidence of Trade-Related Labour-Income Shocks in the UK*

Vahid Ahmadi[†]

July 2026

Abstract

This paper examines how the UK tax–benefit system cushions imposed labour-income losses associated with exposure to the 2025 US tariffs. I link 2024–25 Family Resources Survey households to SIC-level export exposure and use PolicyEngine UK to simulate earnings-equivalent wage cuts, job displacement, inactivity and reallocation. Under full-tariff exposure, the imposed annual earnings stress is £0.886 billion. The system offsets 41.3 per cent under wage cuts and 36.4 per cent on average under displacement; Exchequer costs are £0.495 billion and £0.408 billion. Results use 100 assignments and distinguish assignment dispersion from numerical Monte Carlo error. The exercise is static, partial-equilibrium and first-round. It does not estimate tariff pass-through, production, labour demand, prices or total welfare: the tariff-to-wage-bill bridge and adjustment margins are imposed scenarios. The paper shows how labour-market adjustments reshape statutory fiscal and distributional incidence conditional on a common expected earnings loss.

Keywords: tariffs, trade shocks, microsimulation, poverty, inequality, public finances

JEL codes: F13, F16, H23, I38, D31

[*]**Acknowledgements:** This is an open, reproducible research project. The author thanks PolicyEngine contributors for the underlying microsimulation infrastructure. All interpretations and remaining errors are the author's own.

[†]Research Associate at PolicyEngine. Email: vahid@policyengine.org.

1 Introduction

This paper is a static, partial-equilibrium, first-round stress test of the UK tax–benefit system under alternative labour-income scenarios calibrated to sectoral tariff exposure. It is not a model or estimate of the economic transmission of tariffs. In a complete trade model, tariffs first change border prices and quantities, sourcing and market access; firms then adjust production, intermediate demand, value added, mark-ups, investment and productivity; only subsequently do labour demand, hours, wages, employment and household income respond. This paper does not estimate those links. It replaces them with an imposed reduced-form tariff-to-wage-bill bridge and alternative assumed adjustment margins, then evaluates the statutory tax–benefit consequences conditional on each scenario.

Who bears a foreign tariff shock after the tax–benefit system responds—and how much does the answer depend on whether workers adjust through job loss, wage cuts in place, exit into economic inactivity, or reallocation into lower-paid services work? In April 2025 the United States imposed a 10 per cent baseline tariff on UK goods, with 25 per cent rates on cars and steel; the May 2025 Economic Prosperity Deal (EPD) cut the auto rate to 10 per cent on a 100,000-vehicle quota, offered conditional steel relief, and—in a December 2025 side agreement paid for through NHS pricing concessions—exempted pharmaceuticals. A rich institutional literature has traced the macroeconomic, sectoral, regional and firm-level consequences of this shock (OBR, 2025; Bank of England, 2025; City-REDI, 2025; Centre for Cities, 2025); separate tariff studies map consumer prices and income exposure into heterogeneous household welfare (Artuc et al., 2021; Luu et al., 2020). This paper makes the narrower contribution of a UK statutory tax–benefit stress test of alternative partner-country labour-income scenarios.

The calibration combines the tariff schedule, US export exposure by SIC division, and an export-demand scenario parameter to construct a sector exposure index. The exposure shares are built from ONS and HMRC data. A further wage-bill-incidence assumption maps that index directly to employee earnings. This deliberately strong step suppresses the intervening responses of export prices and quantities, domestic sales, imported inputs, production, productivity, profits and capital; it is not a structural production-side equation. The central scenario sets the demand parameter to 1.0. Sensitivities include an OBR-style lower anchor of 0.4, the former April-outturn ratio of 2.0 as a high case, and a severe value of 3.0. None is treated as an estimated coefficient. The resulting wage-bill stress is imposed on Family Resources Survey workers through four assumed adjustment families: wage cuts, displacement, inactivity among selected older workers, and reallocation into selected service industries at an FRS-calibrated earnings penalty. PolicyEngine UK then computes the conditional disposable-income, poverty, inequality and Exchequer responses. Employment and wages are scenario inputs at the microsimulation stage, not estimated consequences of the tariff.

The central 100-assignment results show substantial assignment dispersion. Under the full schedule, Bernoulli displacement affects 17.2 thousand weighted workers on average, removes £0.882 ± 0.418 billion of earnings, and is cushioned by 36.4 ± 4.9 per cent; the earnings-equivalent-in-expectation wage-cut scenario is cushioned by 41.3 per cent. The corresponding Exchequer costs are £0.408 ± 0.224 billion and £0.495 billion. Relative BHC poverty rises by 0.025 ± 0.017 percentage points under displacement and does not move at displayed precision under wage cuts. Under the EPD schedule, displacement affects 13.5 thousand weighted workers on average, costs the Exchequer £0.312 ± 0.203 billion and raises poverty by 0.020 ± 0.015 points. All $\pm$ values are assignment SDs; numerical Monte Carlo SEs are stored separately in the result artifacts.

Two sectoral nuances discipline the headline claims. Steel carries a caveat: roughly 80 per cent of UK steel exports go to the EU, so the steel–Wales cell is more exposed to EU/UK safeguards than to US tariffs. And pharmaceuticals—exempted under the December 2025 deal in exchange for VPAG rebate cuts and NHS pricing concessions—is a fiscal and pricing channel, not an employment channel, and is treated as such. Finally, an EPD counterfactual (full tariffs versus deal-mitigated rates) prices what the deal was worth, sector by sector and region by region, to households and the Treasury. Following Ignatenko et al. (2025), revenue recycling can flip aggregate employment signs; we compute first-round fiscal incidence holding fiscal policy fixed.

The remainder of the paper proceeds as follows. Section 2 situates the study between the quantitative trade-incidence literature and tax–benefit microsimulation. Section 3 sets out the tariff schedule, the trade-by-SIC exposure mapping, the derived sector shocks and the four adjustment-margin families. Section 4 presents the central cushioning, distributional and fiscal results and the EPD counterfactual. Section 5 discusses the policy interpretation. Section 6 concludes. Exploratory reallocation, outturn-shaped, supply-chain and geographical exercises are reported in the appendix.

2 Literature

This paper sits at the intersection of four literatures: the quantitative and reduced-form work on the incidence of trade shocks; the evidence on how workers actually adjust to them; the institutional analyses of the 2025 US tariffs on the UK; and the microsimulation literature on tax–benefit cushioning of earnings shocks. Existing macro–micro studies impose labour-market shocks on tax–benefit models, while tariff studies map price and income exposure into heterogeneous household welfare. The World Bank's Household Impacts of Tariffs framework combines household consumption and income portfolios (Artuc et al., 2021); the OECD links its METRO trade model to household-budget surveys through an explicit GTAP–COICOP concordance (Luu et al., 2020). The narrower contribution here is a UK application that maps the 2025 partner-country export-demand exposure into alternative labour-income stress scenarios and then through the statutory tax–benefit system. It is a first-round scenario exercise, not a causal estimate of the tariffs' total household incidence.

2.1 Trade shocks and their incidence

Two recent papers anchor the structural end and frame the adjustment-margin question as its poles. Ignatenko et al. (2025) build a quantitative general-equilibrium model—nesting Armington, Eaton–Kortum, Krugman and Melitz structures with exact hat algebra over 194 countries—in which the 2025 tariffs enter as a *primitive* on trade shares. Wages are flexible and labour supply elastic, so employment is an equilibrium *outcome*; strikingly, its sign flips with revenue recycling (a tariff-financed income-tax cut raises US employment by 0.32 per cent, a lump-sum rebate lowers it by 0.41 per cent), and the optimal US tariff is a uniform 19 per cent independent of bilateral deficits. The UK is in their sample, and their replication data provide the UK-specific welfare and employment cells we use as aggregate anchors. Atkin et al. (2025), by contrast, apply a Baqaee–Farhi wedge framework to Chilean administrative data with factor supplies *fixed*: there is no employment margin at all, all adjustment runs through flexible wages and prices, and household incidence arrives via consumption baskets, factor ownership, profits and transfers—their "statutory versus economic incidence" framing is the one we adopt. One paper derives employment from the tariff primitive; the other assumes it fixed and lets wages absorb everything. A third structural anchor sits between the poles: Rodríguez-Clare et al. (2025) embed the 2025 trade war in a dynamic trade-and-reallocation model with *downward nominal wage rigidity*, in which higher tariffs raise US manufacturing employment but lower total employment as falling real wages depress participation, with US real income down about 1 per cent by 2028—rigid wages turn the same primitive into an employment shock that flexible-wage models turn into a wage shock, which is precisely the disagreement our adjustment-margin axis makes an explicit scenario dimension. On the partner-country side, Michelena et al. (2025) run the 2025 escalation through a multiregional input–output framework and find employment and export losses concentrated in the *targeted* countries—the same statutory-versus-economic distinction at the country level, but at the level of sectoral aggregates rather than households. Complementing the model-based work, den Besten and Känzig (2025) construct a narrative series of US tariff shocks over 1840–2024 and find tariff increases robustly contractionary—imports fall sharply, exports decline with a lag, and output drops persistently—historical discipline for treating the shock as a negative export-demand shock rather than a benign relative-price change.

The reduced-form lineage begins with Autor et al. (2013): import competition caused large, persistent employment and wage losses in exposed US commuting zones, with effects persisting to 2019 and no offsetting migration (Autor et al., 2021). Caliendo et al. (2019) provide the structural counterpart—a dynamic spatial model with costly mobility in which the China shock is a productivity/trade-cost primitive, generating aggregate gains alongside concentrated regional losses—and Adão et al. (2023) supply the theory-consistent mapping from regional shift-share estimates to aggregate counterfactuals that cautions against reading regional results as national ones. Kim and Vogel (2021) translate reduced-form employment and wage responses into money-metric worker welfare, finding regressive incidence with displaced workers' losses exceeding wage losses; Galle et al. (2023) show in a Roy-model setting that aggregate gains coexist with group-specific losses. Our contribution relative to this structural incidence work is to replace stylised worker groups and stylised transfers with real households and the actual UK tax–benefit system.

The 2018–19 US tariff episode supplies the sharpest direct evidence on who pays a tariff. Pass-through into US import prices was essentially complete, placing the incidence on the importing country's firms and consumers (Amiti et al., 2019; Fajgelbaum et al., 2020), with retail prices adjusting more slowly than border prices (Cavallo et al., 2021)—a pattern the 2025 wave repeats: matching daily retailer microdata to product-level tariff rates, Cavallo et al. (2025) estimate retail pass-through of about 20 per cent by September 2025, contributing roughly 0.7 percentage points to the US CPI; and Flaaen and Pierce (2019) show that US manufacturing employment fell on net, as input-cost increases and foreign retaliation outweighed import protection. The retaliation side of that episode is the closest analogue to our channel—a partner's tariff arriving as an export-demand shock: Waugh (2019) finds Chinese retaliatory tariffs depressed consumption in exposed US counties, and Blanchard et al. (2019) trace the electoral fallout of retaliatory tariffs across US counties. The broader distributional literature—surveyed in Fajgelbaum and Khandelwal (2016) and Fajgelbaum and Khandelwal (2022), with Borusyak and Jaravel (2021) decomposing expenditure against earnings channels—establishes that trade shocks redistribute along both consumption and income margins. The narrower contribution here is to apply a UK statutory tax–benefit microsimulation to alternative labour-income stress scenarios; it does not claim that the broader trade-and-household intersection is empty. Two propagation channels our direct-channel design deliberately excludes have well-measured analogues in this literature. Firm-level shocks travel along supply chains—Barrot and Sauvagnat (2016) show idiosyncratic disruptions imposing substantial output losses on customers of affected suppliers, and Acemoglu et al. (2016) find network effects on aggregate outcomes several times larger than the direct effect—and tradable-sector job losses can affect local services (Moretti, 2010; Gathmann et al., 2020). These estimates are not mechanically composable with our accounting sensitivity; Section 6 carries the caveat. Open-economy macro adds an offsetting margin: exchange rates absorb part of a tariff, but imperfectly and state-dependently—at most a fifth of the 2018–19 dollar appreciation is attributable to the tariffs (Jeanne and Son, 2024), the dollar appreciates on unilateral tariffs but depreciates under retaliation (Corsetti et al., 2025), and tariff *uncertainty* itself depreciated the dollar in 2025 through a risk-premium channel (Kalemli-Özcan et al., 2026); model-based analyses of retaliation similarly find non-trivial consumption losses for passive trade partners (Auray et al., 2020). We hold the exchange rate fixed; no scenario parameter is claimed to absorb or identify that response. Early assessments of the 2025 wave itself (Fajgelbaum and Khandelwal, 2026) again find the incidence landing on the tariff-imposing country's consumers—complementary to our question, which is the incidence on the tariffed *partner's* workers.

2.2 Adjustment margins: how workers absorb trade shocks

The worker-level evidence says the margin is mostly earnings, not permanent unemployment. Autor et al. (2014) find US workers exposed to import competition lose earnings amid churn across employers and sectors rather than through indefinite joblessness. Traiberman (2019) shows for Denmark that occupation-specific human capital produces reallocation with persistent wage losses; because our shock is an *export-demand* shock rather than import competition, the most direct microeconometric precedents are Hummels et al. (2014), who instrument Danish firm-level export demand and trace it into worker wages by task, and Garin and Silvério (2024), who show that Portuguese exporters pass idiosyncratic export-demand shocks into the wages of incumbent workers rather than shedding them—firm-level evidence for exactly the wage-cut family of our scenario grid. Dauth et al. (2021) provide the cleanest evidence for the wage-margin family, with German workers reallocating into services and other plants at lower wages. Dix-Carneiro (2014) estimates that reallocation takes years, with adjustment costs of 1.4–2.7 times annual wages and older workers hit hardest—calibrating our transition horizons and age gradients—while Dix-Carneiro and Kovak (2017) show regional effects that *grow* over two decades on a joint wage/employment margin.

The UK's own history points to a third margin absent from the US and German studies: benefit-financed economic inactivity. Beatty et al. (2007) document that coalfield job losses became male inactivity on incapacity benefits, not measured unemployment, with incomplete recovery after twenty years; Beatty and Fothergill (2017) show that 1980s–90s industrial job destruction *still* inflates the deficit through higher benefit claims and lower tax receipts—the direct UK precedent for our fiscal-incidence question, retrospective and area-level where ours is prospective, rules-based and household-level. The margin is not merely historical: Roberts and Taylor (2022) show in 2009–18 longitudinal data that disability benefit claims still respond to local labour-market slack over and above health status, Beatty and Fothergill (2023) document the persistence of hidden unemployment among incapacity claimants across large parts of Britain, and the IFS records 4.2 million working-age people on health-related benefits by 2024, up from 3.2 million in 2019 (IFS, 2024)—the inactivity margin is live at exactly the moment the tariff shock arrives. Aragón et al. (2018) add gender-differentiated margins from mine closures, and Rud et al. (2022) decompose displacement costs into wage cuts on re-employment versus non-employment spells—directly calibrating the split between our displacement and wage-cut families. For UK-specific elasticities, De Lyon and Pessoa (2021) is the key source: UK workers in Chinese-import-competing industries lost employment years and earnings without equally gainful re-employment, with larger female losses through employment years (see also Dorn and Levell, 2024). Closest to our object, Irastorza-Fadrique et al. (2025) study UK *household* responses to trade shocks—added-worker effects, later retirement, self-employment—but model behaviour, not tax–benefit mechanics; we treat their margins as complementary to our rules-based incidence.

2.3 The 2025 tariffs and the UK

The institutional literature on the 2025 shock is rich but stops above the household. NIESR's NiGEM scenarios and the OBR's March 2025 tariff analysis provide the macro anchors—the OBR puts the permanent UK GDP loss from a 20-point US tariff at one-third to three-quarters of a per cent (NIESR, 2024; OBR, 2025). The Bank of England estimates the effective US tariff on the UK at roughly 9 per cent (against about 18 per cent globally) and finds trade diversion has mildly *lowered* UK import prices (Bank of England, 2025; Dhingra, 2025)—which is what licenses our isolating the export-demand/earnings channel while holding the price and monetary channels fixed; the Bank's own staff work on the exchange-rate response to tariffs and retaliations (Corsetti et al., 2025) supplies the FX side of that held-fixed channel. The multilateral institutions frame the aggregates: the IMF's October 2025 outlook has global growth slowing to 3.2 per cent in 2025 amid the tariff shock (IMF, 2025), and the WTO's October 2025 update puts merchandise trade volume growth at 2.4 per cent in 2025 but only 0.5 per cent in 2026 as the tariffs bind (WTO, 2025). Firm surveys find most UK firms expecting small negative effects, with the US taking about 22 per cent of UK exports (CEP, 2025). The regional and sectoral geography is sharp: Coventry sends 22 per cent of its exports to the US (Centre for Cities, 2025); auto tariffs are estimated to cost £10bn of GDP over five years with the West Midlands bearing 62 per cent of it (City-REDI, 2025); SMMT data show US vehicle exports collapsing from 7,077 units in April to 4,223 in May 2025 before the EPD's implementation (SMMT, 2025). UK Steel notes that some 80 per cent of UK steel exports go to Europe—the basis of our steel caveat—and the December 2025 pharmaceutical deal (0 per cent tariffs for three years, paid for through VPAG rebate cuts from 22.5 to 14.5 per cent and NHS pricing concessions) makes pharma a fiscal/pricing shock rather than an employment shock (ABPI, 2025). The Resolution Foundation's household-facing analysis is descriptive rather than microsimulated (Resolution Foundation, 2025a); the IFS has no dedicated distributional publication on the 2025 tariffs. The closest work leaves room for a narrower UK statutory tax–benefit stress-test contribution, whose novelty should be rechecked before circulation.

For UK trade-shock precedents more broadly, Dhingra et al. (2017) map Brexit trade costs to income losses, and Fetzer and Wang (2024) provide the best district-level trade-shock incidence estimates (synthetic-control output losses of 5–10 points concentrated in manufacturing districts); Fetzer (2019) and Bloom et al. (2019) supply the political-economy and uncertainty channels. On household prices, Breinlich et al. (2022) use product-level import shares to identify the pass-through of the referendum depreciation, estimating a 2.9 per cent consumer-price increase and a similar real-wage loss. Levell et al. (2017) show why incidence can differ across UK households: food has much greater import content than consumption overall and household budget shares differ. These studies benchmark a consumption-price channel that the present cash-income exercise deliberately holds fixed.

The institutional modelling benchmark is also relevant to what this exercise does and does not claim. The UK Trade Modelling Review recommends a relatively simple, empirically grounded CGE core, complemented by granular partial-equilibrium modules, regional and dynamic extensions, parameter ranges and ex-post validation (Department for International Trade, 2021). In that architecture the tariff is a policy primitive: relative prices and trade quantities adjust first, production and sectoral value added respond next, and factor demand, wages and employment emerge jointly with reallocation across activities. Firm heterogeneity adds selection and productivity responses, while input–output structure transmits intermediate-cost and demand shocks. The OECD METRO model embodies this sequence in a multi-sector, multi-region production system before mapping induced prices into household expenditure and compensating variation (Luu et al., 2020). The World Bank instead offers transparent first-order household accounting of consumer and producer-income exposure, while explicitly excluding substitution, variety and productivity responses that can dominate beyond the short run (Artuc et al., 2021). Relative to those economy-wide or price-side tools, this paper supplies only the terminal tax–benefit module: it conditions on an imposed labour-income distribution rather than deriving it from production, productivity or labour demand.

2.4 Tax–benefit cushioning of earnings shocks

The methodological ancestors are Dolls et al. (2012), who use EUROMOD and TAXSIM to show EU tax–benefit systems absorb on average about 38 per cent of a proportional income shock and 47 per cent of an unemployment shock (against roughly 32 per cent in the US). The EU averages are not the right comparator for this paper; their *UK* cells are, and they are close to ours: the UK income-shock stabilisation coefficient is 0.352 (income tax 0.267, social insurance contributions 0.054, benefits 0.031) and the UK unemployment-shock coefficient 0.415 (tax 0.191, SIC 0.061, out-of-work benefits 0.163). Our 41.3 per cent wage-cut rate and 36.4±4.9 per cent displacement rate are of a similar order, which is a useful benchmark rather than an external validation test. Section 4.2 instead focuses on the *composition* of the stabilisation total and the exposed-worker population. The interpretation of the tax term as an effective *marginal* rate is theirs too—and older still: Auerbach and Feenberg (2000) originate the framing of the income tax as an automatic stabiliser operating at marginal rates, and McKay and Reis (2016) formalise how progressivity and the extensive/intensive structure of shocks determine stabilisation in general equilibrium. The asymmetry between marginal and average relief is therefore not our discovery; our application shows how that asymmetry operates for the selected exposed-sector population. A second ancestor is Sutherland and Figari (2013) on EUROMOD's stress-test use—but both apply stylised macro shocks, not trade-policy-derived ones. Our direct methodological sibling is Brewer and Tasseva (2021), who pass the Covid-19 labour-market shock through UKMOD to show that the UK tax–benefit system plus the emergency support schemes held the rise in income inequality far below the rise in earnings inequality: the same architecture—an externally derived shock realised on survey households and run through the full UK benefit rules—applied to a pandemic where ours applies it to a trade-policy shock, without the emergency schemes and with the adjustment margin as the central axis. The wider macro-micro linkage literature supplies the lineage for the shock-derivation step itself: Bourguignon et al. (2008) codify the techniques for passing macro shocks to household microdata, Peichl (2016) surveys the linkage of microsimulation to structural macro models, and Navicke et al. (2013) show how EUROMOD nowcasts impose estimated labour-market transitions on survey microdata—the same device as our displacement draws, applied there to observed rather than counterfactual shocks. On the US side, exposed commuting zones saw rising disability, retirement and welfare payments that offset only a small fraction of earnings losses (Autor et al., 2013); Hyman et al. (2024) show wage insurance raises re-employment and earnings, a candidate counterfactual reform for this framework. The nearest tariff-microsimulation links are on the US side: the Tax Policy Center's tariff models allocate tariff burdens to tax units through consumption baskets in a full microsimulation model (Tax Policy Center, 2025), the Budget Lab at Yale prices the 2025 schedule's fiscal and distributional effects by income decile (Budget Lab, 2025), and Acosta and Cox (2024) document that the US tariff code is built regressively, with higher rates on lower-end varieties of the same goods. Two institutional facts motivate the cushioning asymmetry examined here. UC's capital rules—thresholds frozen in cash terms since 2006—reduced or extinguished entitlement for around two million income-eligible families, with roughly 1.1 million standing to gain from uprating alone (Resolution Foundation, 2025b); and take-up of income-related benefits is well below 100 per cent (DWP, 2024), so statutory replacement rates can overstate what displaced workers receive. Take-up gates entitlement inside the model and is explored only in a five-assignment sensitivity grid (Section 3.4), while the capital rules are a real-world gate the plain FRS data cannot exercise (Section 6). The distinction between consumer-price incidence and a partner's export-demand/earnings incidence defines the paper's narrower contribution: a UK statutory fiscal stress test organised around alternative adjustment margins. The results do not overturn the automatic-stabiliser literature; they show how the tax and benefit components differ across imposed margins for this exposed-sector population.

3 Methodology

3.1 Design: primitives first

Throughout, "derived" means generated by the stated scenario mapping, not causally estimated. The central demand parameter is a transparent unit stress, $\varepsilon = 1.0$, rather than an estimate. The grid includes $\varepsilon = 0.4$ as an OBR-style lower anchor, 2.0 as the former April-outturn high case, and 3.0 as a severe sensitivity. The April movement is not used to identify the centre because it also reflects anticipation, front-running, prices, exchange rates, composition and other demand conditions. Likewise, $\pi = 1.0$ is a deliberately strong full wage-bill-incidence assumption. The parameter grid is scenario sensitivity, not a statistical confidence interval.

A reader is entitled to ask, before any number is reported, which parts of this design are read off data and which are assumed. The answer, stated in full:

Object	Status
Tariff schedule τ_j	**Data** (policy documents: the April 2025 schedule, the May 2025 EPD, the December 2025 pharmaceutical agreement).
US-export intensity x_j	**Data** (HMRC RTS/OTS customs exports over ONS ABS division turnover; Appendix A).
Export fall	**Data** in the measured family (realised HMRC outturn, May 2025–February 2026); **assumed** in the tariff families through declared low, central, high and severe demand scenarios.
Pass-through π	**Assumed** ($\pi = 1.0$), varied on a grid (Appendix D).
Adjustment margin	**Assumed**, and deliberately varied. This is the paper's free dimension.
Household incidence	**Computed** from statutory tax and benefit rules in PolicyEngine UK; no behavioural or reduced-form step.

The margin is assumed because the literature does not agree on it, and the disagreement is structural rather than empirical: Ignatenko et al. (2025) derive the employment response from elastic labour supply and flexible wages, while Rodríguez-Clare et al. (2025) derive it from downward nominal wage rigidity, so the same tariff primitive arrives as a wage shock in the first model and a job shock in the second. We do not endogenise the margin. We price the disagreement—holding the aggregate earnings loss fixed and reporting what each pole implies for households and the Exchequer. That is a weaker claim than a structural model makes, and a more auditable one.

The shock enters through primitives, not through an assumed employment effect. The pipeline is:

$$s_j \;=\; \tau_j \, x_j \, \varepsilon \, \pi, \tag{1}$$

where s_j is an imposed employee-wage-bill stress for SIC 2007 division j, τ_j is its tariff rate, x_j is US-export intensity, ε is an export-demand calibration and π is a *wage-bill-incidence assumption*. Calling π pass-through would conflate distinct mechanisms. A structural transmission would separately map the tariff into export prices and quantities, quantities into domestic production and value added, and production into factor demand, productivity, profits, wages, hours and employment. Equation (1) compresses those links into one accounting bridge. It neither estimates a production function nor identifies the causal response of productivity, output or labour demand.

x_j is US goods exports as a share of division turnover, built from HMRC trade data and ONS Annual Business Survey turnover denominators; Appendix A documents the frozen provenance and per-division denominator choice. The central $\varepsilon = 1.0$ is a transparent unit stress, not an elasticity estimate. The central $\pi = 1$ assigns the full calibrated exposure to the employee wage bill; it is intentionally severe and is not established by the economics literature. The sensitivity grid varies $\varepsilon\pi$, but this is parameter sensitivity, not an uncertainty interval around an identified coefficient. The implied gross earnings loss is therefore a conditional scenario input. OECD METRO and the UK Trade Modelling Review instead place tariff shocks inside multi-sector production systems in which prices, output, value added and factor demand adjust jointly (Luu et al., 2020; Department for International Trade, 2021); the World Bank HIT framework similarly labels its direct household calculations first-order and excludes productivity and substitution responses (Artuc et al., 2021). This paper begins only after those missing production-side mechanisms would have generated a labour-income distribution.

3.2 The tariff schedule and the EPD counterfactual

Two tariff scenarios bracket the policy. *Full tariffs*: the April 2025 schedule—10 per cent baseline on UK goods, 25 per cent on motor vehicles (SIC 29) and steel (SIC 24). *EPD*: the deal-mitigated schedule—autos at the 10 per cent in-quota rate on up to 100,000 vehicles; steel with conditional, partial relief; pharmaceuticals (SIC 21) exempt under the December 2025 agreement. The automotive scenario applies the 10 per cent rate to the modelled 2024 exposure because the quota is approximately equal to that year's US-bound volume; it does not model quota allocation, above-quota growth or an endogenous volume response around the ceiling. The difference between the two scenarios, run through identical downstream machinery, prices the EPD for households and the Exchequer. Two caveats travel with the schedule. Steel: about 80 per cent of UK steel exports go to the EU, so the US tariff is a minority driver of the sector's exposure and the steel cell should be read alongside EU/UK safeguards. Pharmaceuticals: the exemption was purchased through VPAG rebate cuts and NHS pricing concessions, so pharma's incidence is fiscal and price-side, not an employment shock; we model its employment shock as zero under the EPD and discuss the fiscal channel separately (Section 5).

3.3 Adjustment-margin families

The derived s_j is realised on Family Resources Survey 2024–25 employees in exposed divisions under four central families, plus a factorial mixed-margin stress test. Wage cuts remove $\sum_j s_j B_j$ deterministically; Bernoulli displacement and inactivity remove that amount only in expectation, with realised weighted earnings varying across assignments:

Displacement (ADH short run). Within each exposed division every employee is selected independently with probability s_j, irrespective of the record's survey weight. This Bernoulli design gives each represented worker the intended probability and the sector target in expectation without forcing a weighted head-count quota. Selected workers move to unemployment with earnings, hours, pension contributions and statutory pay zeroed.

Wage cuts with reallocation (Dauth/Traiberman long run). No job loss; every employee in division j takes a proportional cut of s_j, so the weighted aggregate earnings removed equals $\sum_j s_j B_j$ by construction—exact earnings-equivalence with the displacement family's expected removal.

Inactivity (Beatty–Fothergill UK history). The displacement draw, but displaced workers aged 50 and over exit to economic inactivity, the benefit-financed route of UK deindustrialisation, rather than unemployment. The route is benefit-financed in the model, not just relabelled: every inactive worker is flagged as having limited capability for work-related activity, so their household's Universal Credit includes the LCWRA health element (£217.26 a month in 2026). In policyengine-uk 2.89.2 the survey-imputed `uc_LCWRA_element` is stored, so the post-shock benefit-unit value is set to the maximum of the stored amount and one annual statutory element; an existing recipient is never paid a second element, unaffected units are unchanged, and the run hard-errors if the element fails to reach a newly flagged person's benefit unit. This mirrors the Beatty et al. (2007) route by which industrial job loss became incapacity-benefit inactivity. The strong assumption should be stated plainly: *all* displaced workers aged 50 and over are assumed to pass the Work Capability Assessment, so the family is an upper bound on the health-element route; a sensitivity variant in which only 50 per cent of the older displaced qualify is reported alongside (Section 4.4). Work-search conditionality carries no separate fiscal machinery in the model, so the LCWRA element is the entire modelled wedge between inactivity and unemployment.

Reallocation into services (ADH/Autor et al., 2014). The same displacement draw on the same seed—so the two families are paired worker-for-worker—but the drawn workers are re-employed rather than left out of work. Their `sic_industry_division` is reassigned to one of four services destinations (SIC 47 retail, 86 human health, 49 land transport, 56 food and beverage service) in proportion to those divisions' FRS employee-weight shares (29.5, 45.8, 9.2 and 15.5 per cent), and their annual earnings are cut by a penalty calibrated on the FRS itself: weighted mean annual employee earnings of £48,272 over 2,047 hours in the exposed goods divisions against £34,594 over 1,701 hours in the four destinations, a **28.3 per cent annual-earnings penalty** that embeds the hours fall. Two variants are reported: an hours- and age-controlled **low-penalty** case of 14.0 per cent (the pure hourly-wage gap), and a **three-month lag** in which workers are out of work for a quarter of the year before the services job starts, scaling annual earnings and hours by a further $1 - 3/12$. ASHE 2024 full-time medians bracket the hours-controlled figure at 10–20 per cent; the FRS all-employee penalty is larger because it also captures the shift into part-time work. Because this family removes only the penalty on the movers rather than the full earnings-equivalent aggregate, its pound figures are comparable with the displacement family (identical worker sets, paired draws) but *not* with the wage-cut family; its cushioning *rate* is comparable with both. Appendix F reports the results and Appendix B the mechanics.

Mixed adjustment and factorial scenario testing. Let $\lambda \in [0, 1]$ denote the share of the sector shock delivered through displacement. A worker with sector shock s is displaced with probability $p = \lambda s$. Conditional on surviving, the proportional wage cut is

$$c = \frac{s - p}{1 - p},$$

so expected earnings loss is exactly $p + (1 - p)c = s$ for every worker. The endpoints reproduce pure wage cuts ($\lambda = 0$) and pure displacement ($\lambda = 1$). The exploratory grid crosses $\lambda \in \{0, .25, .5, .75, 1\}$ with export-demand calibrations $\varepsilon \in \{0.4, 1, 2, 3\}$ on five common assignments per cell. This follows the companion UK AI study's factorial principle: vary shock magnitude separately from adjustment composition. It is scenario testing, not a probability distribution over future outcomes.

Two scope limits on the family set should be stated here rather than in a footnote. The wage-cut family models the *earnings consequence* of reallocation without moving anyone between sectors: workers keep their observed industry and hours, and the proportional cut stands in for the wage penalty that a completed reallocation would carry; the reallocation family does move them, but who reallocates is assumed (it is the displacement draw) rather than estimated. And no household-side behavioural margin is active—added-worker responses, earlier or later retirement, and shifts into self-employment (Irastorza-Fadrique et al., 2025) are all suppressed, as are any labour-supply responses to the tax and benefit changes themselves. All four families are therefore first-round, rules-based incidence.

The margin is the paper's central axis because the tax–benefit system treats the two poles asymmetrically: an in-work earnings reduction is offset automatically at marginal tax and taper rates, while a job loss moves the worker to the gated, means-tested extensive margin of the benefit system. How much each pole is cushioned, and—the question that turns out to discriminate between them—how much that cushioning depends on anyone doing anything, are empirical questions the microdata answer (Section 4.2).

3.4 Universal Credit take-up after the shock

The take-up experiment is applied symmetrically across margins. After each shock, changed benefit units whose all-claim UC award moves from zero at baseline to positive post-shock receive a claiming draw at the scenario take-up rate. This includes wage-cut households that cross into entitlement. Existing claimants, existing eligible non-claimants, unchanged units and post-shock-ineligible units retain their baseline flag, avoiding an unintended population-wide take-up reform. The grid therefore measures sensitivity to a common new-entitlement convention; it does not estimate actual claiming behaviour.

One modelling choice deserves its own subsection, because an earlier version of this paper got it wrong and the correction changes a headline number. PolicyEngine UK carries benefit take-up as a benefit-unit-level *stored input*, `would_claim_uc`, drawn once at dataset build time. Two properties of that draw matter. First, its target is a flat population-wide parameter—0.55 as a share of *all* benefit units, with reported UC recipients anchored to TRUE and the remainder filled to hit the target. It is not a take-up rate among the entitled, and it is therefore not comparable to the roughly 80 per cent take-up-among-entitled that the Resolution Foundation and the IFS assume; in the dataset used here the realised flag rate is 0.549 over all benefit units and 0.541 among UC-eligible ones. Second, the flag is drawn on *pre-shock* circumstances. Nothing in a naive shock pipeline re-draws it, so a worker displaced by the shock carries into unemployment the claiming decision attributed to them while they were employed and, in the great majority of cases, not entitled to anything—measured take-up among the displaced post-shock was 46.9 per cent, below even the population flag rate. The baseline flag is thus not informative about the post-shock claiming behaviour of a newly entitled family: it encodes a decision taken in a state of the world the shock has destroyed.

The pipeline therefore re-draws `would_claim_uc` after each shock only for benefit units whose circumstances changed and whose all-claim UC award moves from zero at baseline to positive post-shock, at a scenario parameter `uc_takeup` with default 0.80. The re-draw uses an independent random stream, leaving worker assignment unchanged and preserving paired comparisons. This rule applies to displacement, inactivity, reallocation and wage cuts alike; existing eligible, post-shock-ineligible and unchanged units retain their baseline flag. Take-up sensitivity is therefore evaluated symmetrically for newly entitled units, although the parameter remains assumed rather than observed.

3.5 Microsimulation and outcomes

The Monte Carlo draws describe sensitivity to the artificial assignment of displacement across weighted FRS records. Their standard deviations are assignment dispersion, not sampling standard errors, confidence intervals, or uncertainty about the tariff response, survey design or model parameters. Numerical Monte Carlo error of a reported mean is the assignment SD divided by the square root of the number of draws. Parameter scenarios vary the demand calibration, wage-bill incidence, take-up and adjustment composition, but are not assigned probabilities and do not form confidence intervals. The FRS complex-survey sampling variance and uncertainty in the trade-to-earnings bridge are not estimated. Reported comparisons are therefore descriptive scenario contrasts. Distributional cash-income changes use annual household disposable-income change divided by household size before person-weighted aggregation; aggregate disposable-income totals remain household-weighted totals.

The realisation step—an imposed status and earnings transition on survey households, then run through the full benefit rules—follows Brewer and Tasseva (2021), who apply it to the Covid-19 labour-market shock in UKMOD, and the EUROMOD nowcasting tradition of imposed labour-market transitions (Navicke et al., 2013). PolicyEngine UK computes baseline and shocked household disposable income (HBAI cash concept throughout), relative before-housing-costs poverty, absolute before-housing-costs poverty against a fixed baseline line, after-housing-costs poverty, the Gini coefficient, distributional breakdowns and the Exchequer effect. Central direct and measured scenarios use 100 assignments; explicitly exploratory appendix diagnostics use five. Following Ignatenko et al. (2025), revenue recycling can flip aggregate employment responses; here fiscal policy is held fixed. Only direct taxes and benefits respond. Indirect taxes are not shocked, so reduced consumption-tax receipts are absent from the government balance.

4 Results

All central results use FRS 2024–25, period 2026, and 100 assignments. Values after ± are standard deviations across artificial assignments, not standard errors or confidence intervals. Numerical Monte Carlo SE (SD divided by $\sqrt{100}$) is stored separately in each JSON artifact. Five-assignment diagnostics are labelled exploratory.

4.1 Derived sector shocks

The deterministic full-schedule direct earnings stress is £0.886 billion annually; the EPD stress is £0.693 billion. These are scenario inputs generated by Equation (1), not estimated causal effects. The sector rates are exactly half those of the former $\varepsilon = 2$ high case.

4.2 Cushioning

Table 1: Central direct-channel scenarios, 100 assignments

Schedule	Margin	Workers (thousands)	Gross loss (£bn)	Exchequer (£bn)	Cushioning (%)
Full	Displacement	17.2	0.882 ± 0.418	0.408 ± 0.224	36.4 ± 4.9
Full	Wage cut	—	0.886	0.495	41.3
Full	Inactivity	17.2	0.882 ± 0.418	0.419 ± 0.226	37.7 ± 5.1
EPD	Displacement	13.5	0.678 ± 0.374	0.312 ± 0.203	35.7 ± 6.0
EPD	Wage cut	—	0.693	0.386	41.1
EPD	Inactivity	13.5	0.678 ± 0.374	0.320 ± 0.206	37.0 ± 6.3

Notes: Worker, gross-loss and draw-based-margin values are means ± assignment SD. Wage-cut dispersion from the common UC claiming redraw is below the displayed precision. Earnings equivalence holds in expectation, not assignment by assignment.

The full-schedule displacement and wage-cut cushioning rates differ in their means, while displacement retains material assignment dispersion. For the displacement mean, the numerical Monte Carlo SE is 0.5 percentage points for cushioning and £0.022 billion for Exchequer cost. These quantities measure simulation precision only. The decomposition remains economically informative: proportional cuts remove the top slice of earnings and therefore receive relief near marginal tax rates; displacement removes the whole earnings stream and receives relief nearer average rates, with UC providing part of the remainder. These are descriptive model mechanisms rather than estimates of behavioural insurance reliability.

Take-up sensitivity. The regenerated grid applies the same post-shock-entitlement claiming rule to all margins. It uses only five assignments per cell and is exploratory. It shows that displacement responds more visibly to the assumed claiming probability, while wage-cut results move little because relatively few wage-cut households change UC entitlement. No grid difference is treated as statistically identified.

4.3 Mechanism

The regenerated decomposition and gate audit are also five-assignment diagnostics. Income-tax relief is larger on the wage-cut margin than on displacement; UC is more important after displacement. The plain FRS input contains no usable household-capital measure, so the model cannot establish the empirical importance of UC's capital test. New-style JSA is not activated by the imposed transition. These omissions make the gate audit a description of the implemented model, not the full UK out-of-work system.

4.4 Distributional incidence

Under full displacement, relative BHC poverty rises by 0.025 ± 0.017 percentage points. Under the wage-cut scenario, relative BHC poverty is unchanged at displayed precision. The national poverty changes are economically small relative to the uncertainty omitted from the exercise and are not a headline estimate. They can coexist with large losses for selected households because the exposed population is limited and often above the poverty line before the shock.

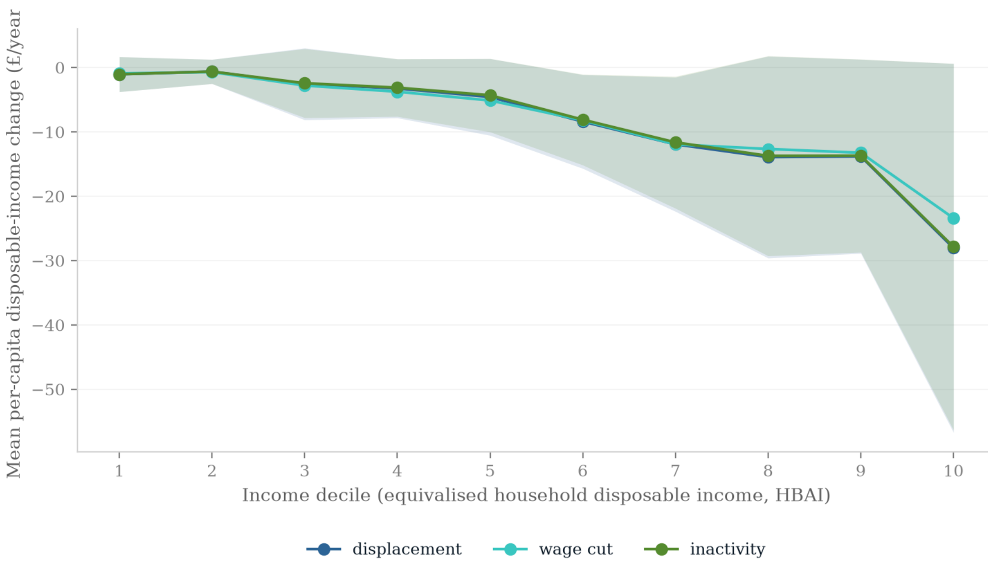

Figure 1: Mean per-capita disposable-income change by baseline equivalised-income decile under the regenerated full-schedule scenarios. Shading shows assignment dispersion, not sampling uncertainty.

4.5 Fiscal incidence

Table 2: Annual Exchequer cost (£billion)

Schedule	Displacement	Wage cut	Inactivity
Full	0.408 ± 0.224	0.495	0.419 ± 0.226
EPD	0.312 ± 0.203	0.386	0.320 ± 0.206

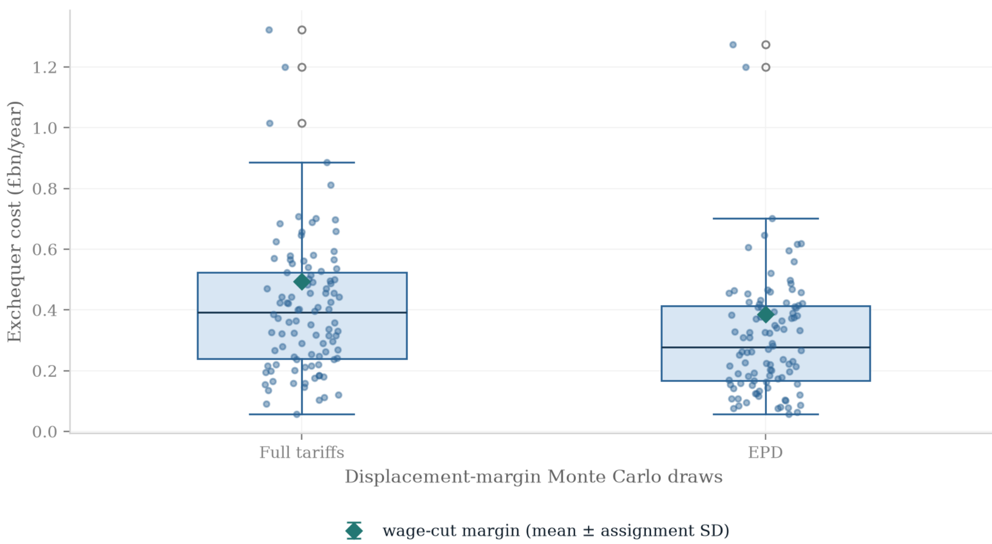

Figure 2: Annual Exchequer cost across regenerated assignments. The spread is assignment dispersion.

4.6 EPD counterfactual

On paired seeds, full minus EPD displacement differs by 3743 ± 2832 workers, £0.204 ± 0.192 billion of gross earnings, and £0.096 ± 0.103 billion of Exchequer cost. The poverty-rate difference is 0.0049 ± 0.0063 percentage points. These are paired scenario contrasts, not causal deal estimates.

5 Policy

5.1 Economic Prosperity Deal

Within the maintained direct-channel scenarios, the paired full-minus-EPD displacement contrast is 3743 ± 2832 workers, £0.204 ± 0.192 billion of gross earnings and £0.096 ± 0.103 billion of annual Exchequer cost. These SDs describe assignment dispersion and the contrasts are not causal estimates of the deal. Autos drive much of the difference; the scenario applies the in-quota rate to fixed exposure without modelling quota allocation or future above-quota trade.

5.2 Pharmaceutical channel

The EPD scenario sets the pharmaceutical employment shock to zero, but this does not make the exemption free. Its VPAG and NHS pricing concessions sit outside the labour-income microsimulation. A complete policy appraisal would cost those health-budget and price effects alongside the direct tax–benefit scenario.

5.3 Cushioning policy

The simulations support a limited conclusion: extensive-margin protection depends more on means-tested eligibility and claiming than the income-tax response to an intensive earnings

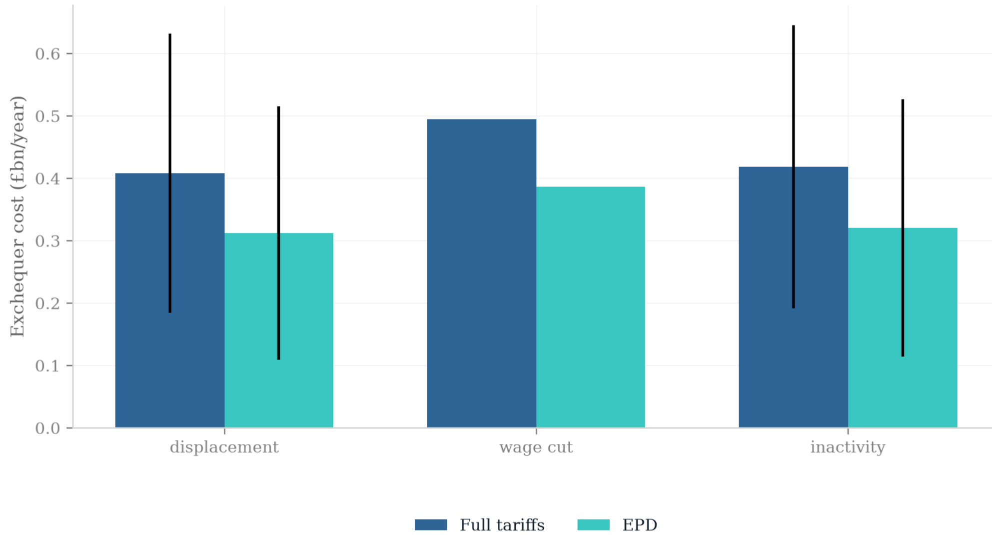

Figure 3: Regenerated full-schedule and EPD fiscal scenarios. Error bars are assignment SDs.

loss. The five-assignment take-up grid is exploratory and does not quantify claimant behaviour precisely. Automatic enrolment after redundancy, contribution-based unemployment insurance and wage insurance are therefore candidate counterfactuals for future analysis, not policy recommendations established by the present results. Their costs and incidence require dedicated simulations and financing assumptions.

The repository also supplies reusable interfaces for the next policy exercises. Observable household and worker characteristics can tilt shock probabilities while preserving each sector's aggregate earnings loss, and a separate price-channel interface converts expenditure-weighted price changes into real disposable income. Neither interface supplies causal coefficients: the former requires externally estimated heterogeneity and the latter requires household expenditure data. They are extensions to populate and validate, not additional headline estimates in this paper.

6 Discussion and conclusion

This paper is a conditional stress test of direct labour-income scenarios, not an estimate of the tariffs' total causal incidence. Under the full schedule, displacement is cushioned by 36.4 ± 4.9 per cent, inactivity by 37.7 ± 5.1 per cent and wage cuts by 41.3 per cent. Exchequer costs are £0.408 ± 0.224 billion, £0.419 ± 0.226 billion and £0.495 billion respectively. Every $\pm$ value is assignment dispersion; numerical Monte Carlo SE is stored separately.

The component mechanism is consistent with automatic-stabiliser theory: proportional earnings cuts receive tax relief near marginal rates, while a whole-job loss receives relief nearer average rates and relies more on UC. The corrected common claiming rule applies to every margin. Five-assignment take-up and gate exercises are exploratory and do not establish behavioural reliability or precise gate shares.

National poverty effects are small in the central scenarios: $+0.025 \pm 0.017$ percentage points under full displacement and zero at displayed precision under wage cuts. These changes should not be interpreted as precise population estimates because survey and production-side uncertainty are omitted. They also do not make selected households' losses small: the exposed population is limited and often above the poverty line before the shock.

The paired full-minus-EPD displacement contrast is 3743 ± 2832 workers and £0.096 ± 0.103 billion of annual Exchequer cost. This is a scenario comparison driven substantially by autos, not a causal evaluation of the deal.

The main limitation is structural: the model starts at the end of the production-side transmission mechanism. It does not estimate how the tariff changes border prices, export quantities, domestic output, value added, imported-input costs, firm productivity, investment, profits or labour demand. It assigns calibrated exposure directly to the wage bill and then assumes how that loss is divided among wages, employment, inactivity and reallocation. The results therefore answer a conditional fiscal-incidence question—what statutory taxes and transfers do after specified labour-income changes—not what the tariff does to the economy. Assignment is also independent within broad SIC divisions; the inactivity scenario assumes WCA passage for selected older workers; New Style JSA and within-year timing are absent; household labour-supply responses are suppressed; and indirect taxes, exchange rates, market switching, retaliation and general equilibrium are outside the central model. The supply-chain calculation is an accounting sensitivity, not a lower bound, and cannot be mechanically combined with local-employment multipliers. Constituency results are illustrative because local industry is not observed at the required level.

The companion UK AI study suggests useful extensions to the analysis architecture: separate the aggregate shock size from its incidence pattern as factorial scenario axes; add mixed wage–employment adjustment rather than only polar margins; benchmark exposure against external employment data; distinguish literature-calibrated from measured-incidence scenarios; evaluate policy counterfactuals on common draws; and generate paper values from a declared build manifest. Those extensions improve transparency and robustness, but they do not replace the missing trade-to-production first stage. The priority for a substantive extension is therefore a product- or firm-level model linking tariffs to prices, quantities, output, value added and productivity before any household microsimulation is run.

The defensible conclusion is therefore modest. For a given direct earnings loss, the adjustment margin materially changes the mix of tax relief, benefit support, poverty and fiscal cost. Quantifying the realised tariff effect requires causal trade and linked employer–worker evidence beyond this microsimulation.

Data and code availability

The analysis code, configuration, tests, input manifest, and generated result artifacts are available in the accompanying GitHub repository. The manifest records source URLs, retrieval dates, vintages, exclusions, and SHA–256 hashes. The Family Resources Survey and certain ONS supply-chain workbooks are licensed or access-controlled and are therefore not redistributed. A fully fresh reproduction requires those files (or the documented cached inputs), the Python environment specified by `uv.lock`, and the commands in the repository's `Makefile`. Continuous integration validates the public code path and manifest; it cannot validate inputs that cannot legally be distributed.

Declarations

The author reports no conflicts of interest. No external funding was received for this study. The project uses secondary, de-identified survey and administrative-source inputs; no new human-participant data were collected. The author is responsible for the analysis, interpretation, and any remaining errors.

References

ABPI (2025). UK–US agreement on pharmaceutical tariffs and pricing. Association of the British Pharmaceutical Industry, London, December 2025.

Acemoglu, D., Akcigit, U. and Kerr, W. (2016). Networks and the macroeconomy: An empirical exploration. *NBER Macroeconomics Annual*, 30(1), 273–335.

Acosta, M. and Cox, L. (2024). The regressive nature of the US tariff code: Origins and implications. Working paper, February 2024 (conditionally accepted, *The Quarterly Journal of Economics*).

Adão, R., Arkolakis, C. and Esposito, F. (2023). General equilibrium effects in space: Theory and measurement. NBER Working Paper No. 25544, National Bureau of Economic Research, Cambridge, MA.

Amiti, M., Redding, S. J. and Weinstein, D. E. (2019). The impact of the 2018 tariffs on prices and welfare. *Journal of Economic Perspectives*, 33(4), 187–210.

Aragón, F. M., Rud, J. P. and Toews, G. (2018). Resource shocks, employment, and gender: Evidence from the collapse of the UK coal industry. *Labour Economics*, 52, 54–67.

Artuc, E., Porto, G. and Rijkers, B. (2021). Household impacts of tariffs: Data and results from agricultural trade protection. *The World Bank Economic Review*, 35(3), 563–585.

Auerbach, A. J. and Feenberg, D. (2000). The significance of federal taxes as automatic stabilizers. *Journal of Economic Perspectives*, 14(3), 37–56.

Atkin, D., Bernadac, V., Donaldson, D., Garg, T. and Huneeus, F. (2025). The incidence of distortions. Working paper, December 2025 draft.

Auray, S., Devereux, M. B. and Eyquem, A. (2020). Trade wars, currency wars. NBER Working Paper No. 27460, National Bureau of Economic Research, Cambridge, MA.

Autor, D. H., Dorn, D. and Hanson, G. H. (2013). The China syndrome: Local labor market effects of import competition in the United States. *American Economic Review*, 103(6), 2121–2168.

Autor, D. H., Dorn, D., Hanson, G. H. and Song, J. (2014). Trade adjustment: Worker-level evidence. *The Quarterly Journal of Economics*, 129(4), 1799–1860.

Autor, D. H., Dorn, D. and Hanson, G. H. (2021). On the persistence of the China shock. *Brookings Papers on Economic Activity*, Fall 2021, 381–447.

Bank of England (2025). *Monetary Policy Report*, May, August and November 2025 issues. Bank of England, London.

Barrot, J.-N. and Sauvagnat, J. (2016). Input specificity and the propagation of idiosyncratic shocks in production networks. *The Quarterly Journal of Economics*, 131(3), 1543–1592.

Beatty, C. and Fothergill, S. (2017). The impact on welfare and public finances of job loss in industrial Britain. *Regional Studies, Regional Science*, 4(1), 161–180.

Beatty, C., Fothergill, S. and Powell, R. (2007). Twenty years on: Has the economy of the UK coalfields recovered? *Environment and Planning A*, 39(7), 1654–1675.

Beatty, C. and Fothergill, S. (2023). The persistence of hidden unemployment among incapacity claimants in large parts of Britain. *Local Economy*, doi:10.1177/02690942231184815.

Blanchard, E. J., Bown, C. P. and Chor, D. (2019). Did Trump's trade war impact the 2018 election? NBER Working Paper No. 26434, National Bureau of Economic Research, Cambridge, MA.

Bloom, N., Bunn, P., Chen, S., Mizen, P., Smietanka, P. and Thwaites, G. (2019). The impact of Brexit on UK firms. NBER Working Paper No. 26218, National Bureau of Economic Research, Cambridge, MA.

Borusyak, K. and Jaravel, X. (2021). The distributional effects of trade: Theory and evidence from the United States. NBER Working Paper No. 28957, National Bureau of Economic Research, Cambridge, MA.

Bourguignon, F., Bussolo, M. and Pereira da Silva, L. A. (eds.) (2008). *The Impact of Macroeconomic Policies on Poverty and Income Distribution: Macro-Micro Evaluation Techniques and Tools*. Palgrave Macmillan and World Bank, Washington, DC.

Brewer, M. and Tasseva, I. V. (2021). Did the UK policy response to Covid-19 protect household incomes? *The Journal of Economic Inequality*, 19(3), 433–458.

Breinlich, H., Leromain, E., Novy, D. and Sampson, T. (2022). The Brexit vote, inflation and UK living standards. *International Economic Review*, 63(1), 63–93.

The Budget Lab (2025). Where we stand: The fiscal, economic, and distributional effects of all U.S. tariffs enacted in 2025. The Budget Lab at Yale, New Haven, CT.

Caliendo, L., Dvorkin, M. and Parro, F. (2019). Trade and labor market dynamics: General equilibrium analysis of the China trade shock. *Econometrica*, 87(3), 741–835.

Cavallo, A., Gopinath, G., Neiman, B. and Tang, J. (2021). Tariff pass-through at the border and at the store: Evidence from US trade policy. *American Economic Review: Insights*, 3(1), 19–34.

Cavallo, A., Llamas, P. and Vazquez, F. (2025). Tracking the short-run price impact of U.S. tariffs. NBER Working Paper No. 34496, National Bureau of Economic Research, Cambridge, MA.

Centre for Economic Performance (2025). UK firms and the 2025 US tariffs: Evidence from the Decision Maker Panel. VoxEU column, Centre for Economic Policy Research, and CEP "in brief" 11639, Centre for Economic Performance, LSE, London. https://cepr.org/voxeu/columns/impact-2025-us-tariff-announcements-uk-firms.

Centre for Cities (2025). The uneven storm: How US tariffs hit UK cities differently. Centre for Cities, London.

City-REDI (2025). The regional impact of US automotive tariffs on the UK economy. City-REDI, University of Birmingham.

Corsetti, G., Lloyd, S. and Ostry, D. (2025). Trading blows: The exchange-rate response to tariffs and retaliations. Staff Working Paper No. 1,139, Bank of England, London.

Dauth, W., Findeisen, S. and Suedekum, J. (2021). Adjusting to globalization in Germany. *Journal of Labor Economics*, 39(1), 263–302.

De Lyon, J. and Pessoa, J. P. (2021). Worker and firm responses to trade shocks: The UK–China case. *European Economic Review*, 133, 103678.

Department for International Trade (2021). *Trade Modelling Review Expert Panel: Report and Recommendations*. Department for International Trade, London.

den Besten, T. and Känzig, D. R. (2025). The macroeconomic effects of tariffs: Evidence from U.S. historical data. NBER Working Paper No. 34852, National Bureau of Economic Research, Cambridge, MA.

Dhingra, S. (2025). Tariffed with the same brush. Speech, Bank of England, October 2025.

Dhingra, S., Ottaviano, G., Sampson, T. and Van Reenen, J. (2017). The costs and benefits of leaving the EU: Trade effects. *Economic Policy*, 32(92), 651–705.

Dix-Carneiro, R. (2014). Trade liberalization and labor market dynamics. *Econometrica*, 82(3), 825–885.

Dix-Carneiro, R. and Kovak, B. K. (2017). Trade liberalization and regional dynamics. *American Economic Review*, 107(10), 2908–2946.

Dolls, M., Fuest, C. and Peichl, A. (2012). Automatic stabilizers and economic crisis: US vs. Europe. *Journal of Public Economics*, 96(3–4), 279–294.

Dorn, D. and Levell, P. (2024). Labour market impacts of the China shock: Why the tide of globalisation did not lift all boats. *Labour Economics*, 91, 102629.

Department for Work and Pensions (2024). Income-related benefits: Estimates of take-up. Official statistics series, DWP, London.

Fajgelbaum, P. D. and Khandelwal, A. K. (2016). Measuring the unequal gains from trade. *The Quarterly Journal of Economics*, 131(3), 1113–1180.

Fajgelbaum, P. D. and Khandelwal, A. K. (2022). The economic impacts of the US–China trade war. *Annual Review of Economics*, 14, 205–228.

Fajgelbaum, P. D. and Khandelwal, A. K. (2026). Tariffs in 2025: Short-run impacts on the U.S. economy. *Brookings Papers on Economic Activity*, Spring 2026 conference draft.

Fajgelbaum, P. D., Goldberg, P. K., Kennedy, P. J. and Khandelwal, A. K. (2020). The return to protectionism. *The Quarterly Journal of Economics*, 135(1), 1–55.

Fetzer, T. (2019). Did austerity cause Brexit? *American Economic Review*, 109(11), 3849–3886.

Fetzer, T. and Wang, S. (2024). Measuring the regional economic cost of Brexit: Evidence up to 2019. CAGE Working Paper No. 1280, University of Warwick.

Flaaen, A. and Pierce, J. R. (2019). Disentangling the effects of the 2018–2019 tariffs on a globally connected U.S. manufacturing sector. Finance and Economics Discussion Series 2019-086, Board of Governors of the Federal Reserve System, Washington, DC.

Galle, S., Rodríguez-Clare, A. and Yi, M. (2023). Slicing the pie: Quantifying the aggregate and distributional effects of trade. *The Review of Economic Studies*, 90(1), 331–375.

Garin, A. and Silvério, F. (2024). How responsive are wages to firm-specific changes in labour demand? Evidence from idiosyncratic export demand shocks. *The Review of Economic Studies*, 91(3), 1671–1710.

Gathmann, C., Helm, I. and Schönberg, U. (2020). Spillover effects of mass layoffs. *Journal of the European Economic Association*, 18(1), 427–468.

Hummels, D., Jørgensen, R., Munch, J. and Xiang, C. (2014). The wage effects of offshoring: Evidence from Danish matched worker–firm data. *American Economic Review*, 104(6), 1597–1629.

Hyman, B., Kovak, B. K. and Leive, A. (2024). Wage insurance for displaced workers. NBER Working Paper No. 32464, National Bureau of Economic Research, Cambridge, MA.

Institute for Fiscal Studies (2024). Health-related benefit claims post-pandemic: UK trends and global context. IFS Report, Institute for Fiscal Studies, London, September 2024.

International Monetary Fund (2025). *World Economic Outlook: Global Economy in Flux, Prospects Remain Dim*. IMF, Washington, DC, October 2025.

Ignatenko, A., Lashkaripour, A., Macedoni, L. and Simonovska, I. (2025). Making America great again? The economic impacts of Liberation Day tariffs. *Journal of International Economics*, 157, 104138.

Irastorza-Fadrique, A., Levell, P. and Parey, M. (2025). Household responses to trade shocks. *Journal of International Economics*, forthcoming (IFS Working Paper 24/52).

Jeanne, O. and Son, J. (2024). To what extent are tariffs offset by exchange rates? *Journal of International Money and Finance*, 142.

Kalemli-Özcan, Ş., Soylu, C. and Yildirim, M. A. (2026). Global trade, tariff uncertainty and the U.S. dollar. NBER Working Paper No. 34728, National Bureau of Economic Research, Cambridge, MA.

Kim, R. and Vogel, J. (2021). Trade shocks and labor market adjustment. *American Economic Review: Insights*, 3(1), 115–130.

Levell, P., O'Connell, M. and Smith, K. (2017). The exposure of households' food spending to tariff changes and exchange rate movements. IFS Briefing Note BN213, Institute for Fiscal Studies, London.

Luu, N., Woloszko, N., Causa, O., Arriola, C., van Tongeren, F. and Johansson, Å. (2020). Mapping trade to household budget survey: A conversion framework for assessing the distributional impact of trade policies. OECD Trade Policy Papers No. 244, OECD Publishing, Paris.

McKay, A. and Reis, R. (2016). The role of automatic stabilizers in the U.S. business cycle. *Econometrica*, 84(1), 141–194.

Michelena, G., Ernst, C. and Bertin, P. (2025). Tariffs and labor markets: The employment impact of the recent trade conflict. A multiregional input–output analysis. International Labour Organization / arXiv:2512.11578, December 2025.

Moretti, E. (2010). Local multipliers. *American Economic Review: Papers & Proceedings*, 100(2), 373–377.

Navicke, J., Rastrigina, O. and Sutherland, H. (2013). Nowcasting indicators of poverty risk in the European Union: A microsimulation approach. EUROMOD Working Paper No. EM11/13, ISER, University of Essex.

NIESR (2024). Implications of higher US tariffs for the UK economy. National Institute of Economic and Social Research, London, December 2024.

Office for Budget Responsibility (2025). *Economic and Fiscal Outlook*, March 2025 (tariff scenario box). OBR, London.

Peichl, A. (2016). Linking microsimulation and CGE models. *International Journal of Microsimulation*, 9(1), 167–174.

Resolution Foundation (2025a). Trump tariff turmoil. Resolution Foundation, London, April 2025.

Resolution Foundation (2025b). Saving penalties: Reforming the capital rules in Universal Credit. Resolution Foundation, London, April 2025.

Roberts, J. and Taylor, K. (2022). New evidence on disability benefit claims in Britain: The role of health and the local labour market. *Economica*, 89(353), 131–160.

Society of Motor Manufacturers and Traders (2025). Written submission to the House of Commons Business and Trade Committee inquiry on UK–US trade (reference UST0041). UK Parliament written evidence, https://committees.parliament.uk/writtenevidence/167352/html/.

Rodríguez-Clare, A., Ulate, M. and Vasquez, J. P. (2025). The 2025 trade war: Dynamic impacts across U.S. states and the global economy. NBER Working Paper No. 33792, National Bureau of Economic Research, Cambridge, MA.

Rud, J. P., Simmons, M., Toews, G. and Aragón, F. (2022). Job displacement costs of phasing out coal. IZA Discussion Paper No. 15581.

Sutherland, H. and Figari, F. (2013). EUROMOD: The European Union tax–benefit microsimulation model. *International Journal of Microsimulation*, 6(1), 4–26.

Tax Policy Center (2025). Description of the Tax Policy Center's tariff models. Urban–Brookings Tax Policy Center, Washington, DC.

Traiberman, S. (2019). Occupations and import competition: Evidence from Denmark. *American Economic Review*, 109(12), 4260–4301.

Waugh, M. E. (2019). The consumption response to trade shocks: Evidence from the US–China trade war. NBER Working Paper No. 26353, National Bureau of Economic Research, Cambridge, MA.

World Trade Organization (2025). *Global Trade Outlook and Statistics: Update October 2025*. WTO, Geneva.

A Data appendix

US exports use HMRC uktradeinfo OTS/RTS data for 2024, mapped from SITC to SIC 2007 divisions and divided by the latest non-suppressed ONS Annual Business Survey turnover. Of a £55.6 billion customs-basis total, £52.5 billion maps to divisions after documented exclusions, chiefly non-monetary gold and precious metals. The measured window uses May 2025–February 2026 against the mean of the same calendar months one and two years earlier. Its 12.7 per cent aggregate fall is descriptive, not causal.

B Shock mechanics

Each employee in exposed division j is independently selected with probability s_j. Weighted displaced counts and earnings therefore meet targets in expectation rather than exactly. Selected workers have employment income, hours, pension contributions and statutory pay zeroed. Wage cuts remove s_j of each exposed worker's earnings deterministically. Reallocation uses the same selected workers, assigns a services destination, and applies the calibrated earnings penalty.

UC claiming is redrawn after every margin for benefit units whose circumstances changed and that are entitled post-shock. The LCWRA override sets the benefit-unit element to the maximum of its stored baseline value and one statutory annual element, preventing duplicate payment.

C Monte Carlo and measures

Central direct and measured scenarios use 100 assignments. Every displayed ± value is an assignment SD, not a standard error or confidence interval. The machine-readable artifacts additionally store numerical Monte Carlo standard error as assignment SD divided by $\sqrt{n}$; it measures precision of the simulated mean only. Survey-design, tariff-response and model-parameter uncertainty are not estimated. Five-assignment take-up, mechanism, demographic, poverty-gap, reallocation, duration, sensitivity-grid and supply-chain exercises are exploratory and are excluded from the headline evidence. The mixed-adjustment scenario surface likewise uses five common assignments per cell. Its machine-readable cells and draws are in `results/scenario_testing.json` and `results/scenario_testing.csv`.

Absolute BHC poverty uses 60 per cent of the baseline equivalised median as a fixed line. The cushioning rate is $1 - \Delta Y / \Delta E$, where ΔE is gross employment-earnings loss and ΔY is household disposable-income loss.

D Elasticity and pass-through grid

The regenerated grid spans $\varepsilon \in \{1, 1.5, 2, 3\}$ and $\pi \in \{0.5, 0.75, 1\}$. Because only five displacement assignments are used per cell, it is an exploratory scaling diagnostic. Equal products $\varepsilon\pi$ generate equal deterministic wage-cut shocks; Bernoulli displacement outcomes vary across assignments. The grid is not an uncertainty interval.

E Factorial adjustment scenarios

Figure 4 separates the aggregate export-demand calibration from the assumed composition of labour-market adjustment. Each cell uses the full tariff schedule and five common assignments. The wage-cut endpoint is stable: cushioning is 40.6–41.3 per cent over the four calibrations. Moving towards displacement generally lowers cushioning, but the small-shock cells are especially noisy because a few high-weight FRS records dominate five Bernoulli assignments. These cells are stress scenarios, not estimates or confidence regions. The qualitative result is that fiscal cost scales mainly with the imposed gross shock, while the adjustment mix changes the share absorbed by taxes and transfers.

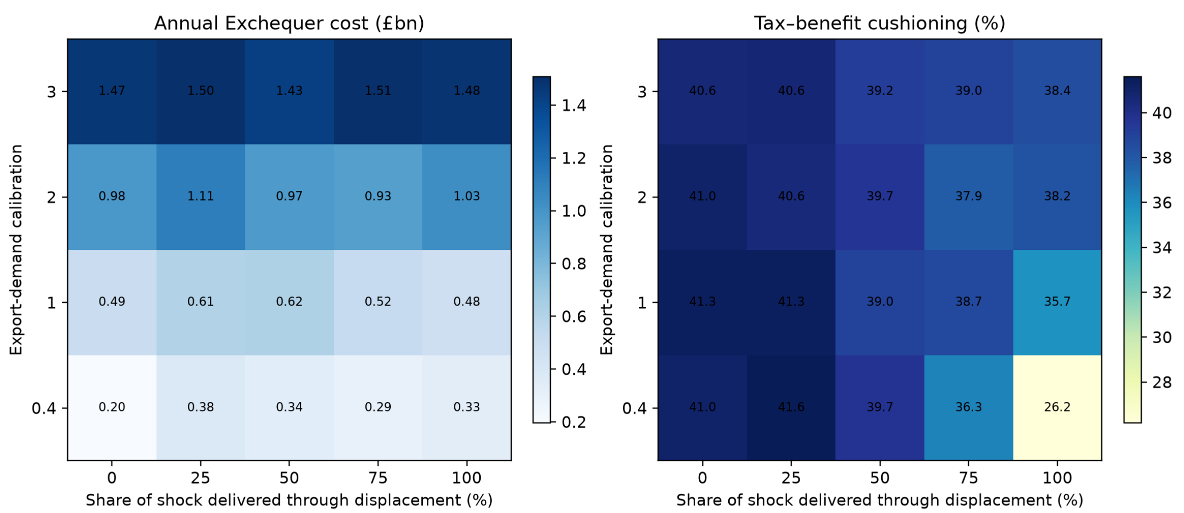

Figure 4: Factorial scenario surface: export-demand calibration by the share of expected sector loss delivered through displacement rather than survivor wage cuts. Cell values are means over five common assignments. Assignment noise, survey uncertainty and uncertainty in the production-side transmission are not combined.

F Reallocation sensitivity

The reallocation exercise uses five paired assignments and is exploratory. Under the full schedule, the central 28.3 per cent penalty produces a mean gross loss of £0.264 billion, Exchequer cost of £0.149 ± 0.131 billion, cushioning of 39.7 ± 3.4 per cent, and zero measured BHC-poverty change. With a three-month lag, gross loss rises to £0.432 billion and the Exchequer cost to £0.245 ± 0.216 billion. The 14 per cent penalty gives gross loss of £0.131 billion and Exchequer cost of £0.075 ± 0.064 billion. These pound levels are not commensurable with the economy-wide wage-cut scenario because only selected movers' penalty is removed.

G Observed-outturn stress scenario

The descriptive May 2025–February 2026 export window falls 12.7 per cent against its two-year same-month baseline. This is not a causal tariff estimate: it includes prices, exchange rates, front-running, rerouting and other demand conditions.

The downside displacement scenario affects 17.3 thousand weighted workers on average, is cushioned by 36.7 ± 5.2 per cent and costs the Exchequer £0.420 ± 0.233 billion. The wage-cut scenario is cushioned by 41.6 per cent and costs £0.528 billion. This is an observed-outturn downside stress case, not a tariff-caused or preferred estimate; the validation artifact also reports signed net exposure before expanding divisions are clipped.

H Demographic diagnostic

The gender, age and household-type breakdown uses five Bernoulli assignments and is exploratory. Selected exposed-sector workers are disproportionately men and concentrated in prime and older working ages, but individual subgroup estimates are too assignment-sensitive for precise rankings. The demographic artifact is therefore a diagnostic rather than an inferential table.

I Duration scope

The model reports the internally coherent full-year endpoint only. An earlier six-month sensitivity combined half-year earnings with full-period unemployment status and was removed because an annual model cannot reproduce within-year tax, waiting-period or eligibility timing. A shorter-duration estimate requires monthly simulations; no such result is claimed here.

J Supply-chain sensitivity

The input–output scenario adds upstream requirements using the ONS domestic Leontief inverse. The accounting shock has a 1.95 direct-plus-upstream factor. It is not a lower bound and is not multiplied by local-employment multipliers. In the exploratory five-assignment microsimulation, displacement affects 29,700 weighted workers on average, costs the Exchequer £0.68 ± 0.31 billion and raises relative BHC poverty by 0.028 percentage points. The wage-cut variant costs £0.96 billion and leaves measured relative poverty unchanged. These figures are accounting sensitivities, not all-channel causal estimates.

K Constituency illustration

The enhanced-FRS constituency layer uses imputed industries rather than observed local industry structure. It is averaged over 20 paired worker-assignment draws (seeds 0–19), replacing the unstable single-seed surface used in an earlier draft. The mapping now divides each household's disposable-income change by household size before constructing constituency means. The CSV reports both the draw mean and assignment-draw standard deviation. Because the constituency weights contain no observed local industry mix and the SIC imputation is held fixed, these remain synthetic demographic and income projections, not constituency exposure estimates; individual seat rankings are therefore deliberately not reported.

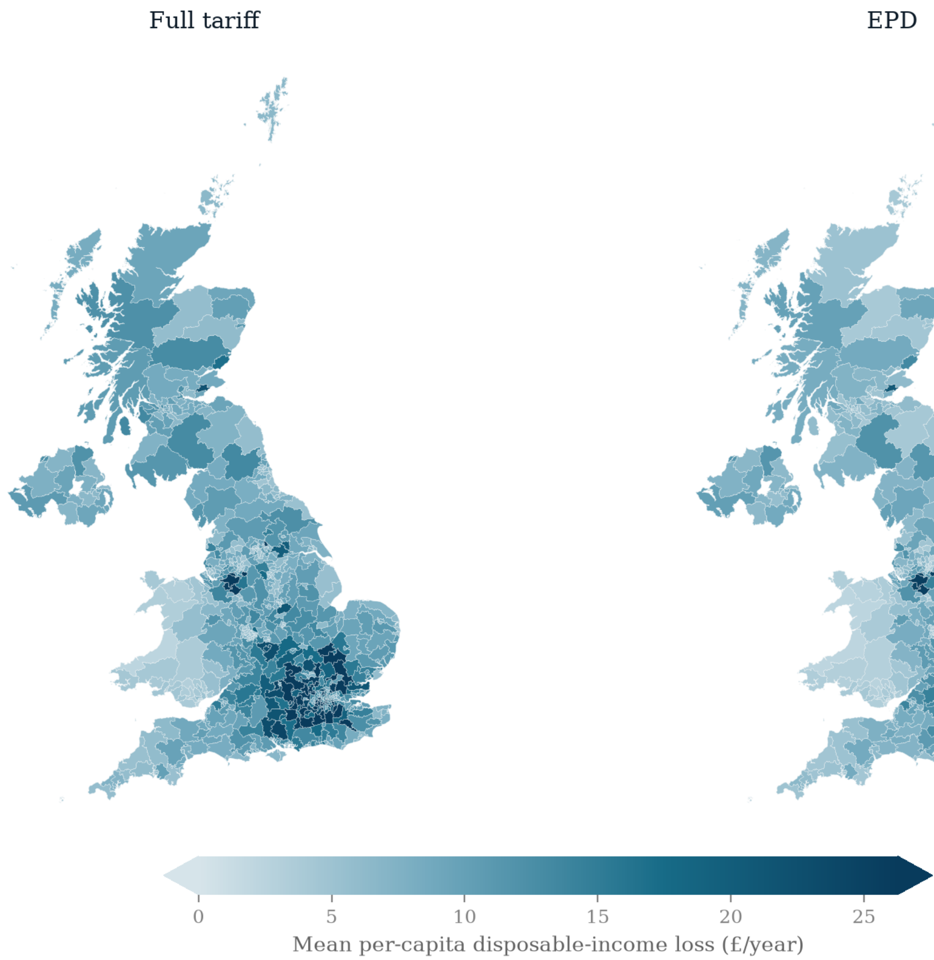

Figure 5: Synthetic constituency illustration of annual per-capita disposable-income loss, averaged over 20 paired worker-assignment draws. A shared sequential scale is clipped at the 95th percentile. The map uses imputed SIC and no observed constituency industry mix; it is not a map of local tariff exposure.